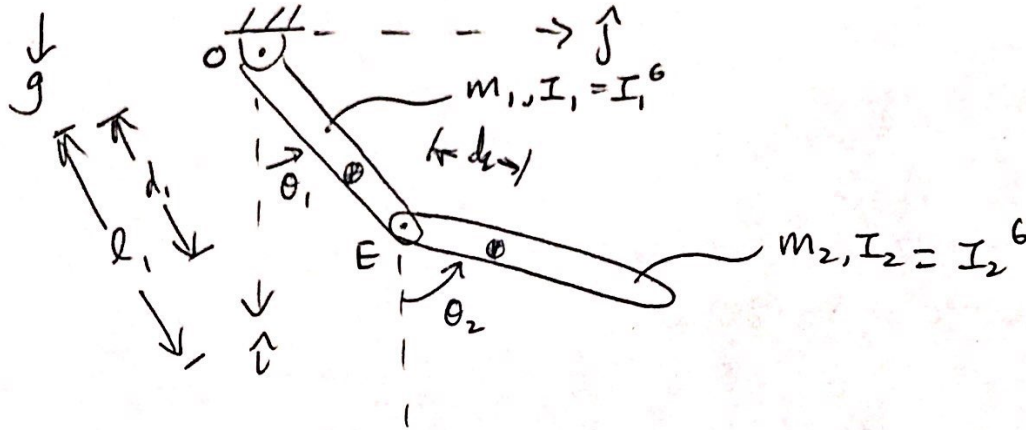


Double Pendulum Derivation using Matlab:

System $AMB/0$ + Forearm AMB/E

- 2 equations in $\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2, \ddot{\theta}_1, \ddot{\theta}_2$
linear in $\ddot{\theta}_1, \ddot{\theta}_2$

$$\Rightarrow \underbrace{\begin{bmatrix} \square & \square \\ \square & \square \end{bmatrix}}_M \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} = \begin{bmatrix} \square \\ \square \end{bmatrix}$$

= stuff w/ no $\ddot{\theta}_1, \ddot{\theta}_2$